

ECON 351 · Labor-Leisure Problems

All practice problems + drills to solve

Sources checked (uploaded course materials):

- Homework 1 Ch.4: No labor-leisure exercises
- Homework 2 Ch.9: No labor-leisure (Ch.9 uncertainty only)
- Sample Questions Ch.4: No labor-leisure questions found
- Sample Questions Ch.9: No labor-leisure questions found
- Mock Midterm 1: No labor-leisure questions found
- Econ351 Lecture (Ch.9 upload): Chapter 9 only · no labor-leisure
- Practice Tests 1-3: Q3 & Q4 · Maria problems (included below)

6 problems in this PDF (2 from practice tests + 4 extra drills).

Labor-Leisure Formula Sheet

Time: $T = L + \ell$ (total hours = labor + leisure)

Budget: $c = w \cdot L + V = w(T - \ell) + V$

Optimum: $MU_{\ell} / MU_c = w$

Cobb-Douglas $U = c^{\alpha} \ell^{1-\alpha}$: at optimum $\ell \cdot c = \alpha \cdot w \cdot T$

Special case $\alpha = 1 - \alpha = 0.5$: $c = w \cdot \ell$ and $\ell^* = (w \cdot T + V) / (2w)$

Labor hours: $L^* = T - \ell^*$

Consumption: $c^* = w \cdot \ell^* + V$

Problem 1: LL-1

Practice Tests 1, 2, 3 · Question 3

Maria has 16 hours per day for leisure ($\cdot$) and work.

Wage $w = \$20/\text{hr}$, non-labor income $V = \$80$.

Utility $U(c, \cdot) = c^{0.5} \cdot^{0.5}$, where $c = w(16 \cdot \cdot) + V$.

- Write the budget constraint.
- Find optimal leisure l^* .
- Find labor hours L^* and consumption c^* .

Show your work below:

Problem 2: LL-2

Practice Tests 1, 2, 3 · Question 4

Same setup as LL-1, but wage rises to $w = \$30/\text{hr}$.

- (a) Find new $\cdot^*$, L^* , and c^* .
- (b) Does Maria work more or less than when $w = \$20$? Explain.

Show your work below:

Problem 3: LL-3

Extra drill · standard Cobb-Douglas

Alex has $T = 8$ hours/day, $w = \$25/\text{hr}$, $V = \$0$.

$$U(c, \cdot) = c^{0.5} \cdot^{0.5}, c = w(8 \cdot \cdot).$$

Find $\cdot^*$, L^* , and c^* .

Show your work below:

Problem 4: LL-4

Extra drill · non-labor income change

Same as LL-3 but $V = \$40$ (lottery winnings).

Find $\cdot^*$, L^* , c^* . Did leisure increase compared to LL-3?

Show your work below:

Problem 5: LL-5

Extra drill · leisure-loving preferences

Jordan: $T = 24$ hrs, $w = \$15/\text{hr}$, $V = \$60$.

$U(c, \cdot) = c^{0.25} \cdot^{0.75}$ (more leisure-loving than Maria).

Use $MU_x/MU_y = w$ to find x^* , L^* , c^* .

Show your work below:

Problem 6: LL-6

Extra drill · wage comparative statics

Sam: $T = 10$ hrs, $V = \$100$, $U(c, \cdot) = c^{0.5} \cdot^{0.5}$.

(a) If $w = \$10$, find $\cdot^*$ and L^* .

(b) If $w = \$40$, find $\cdot^*$ and L^* .

(c) As wage rises, does Sam work more or less?

Show your work below:

Answer Key

LL-1:

BC: $c = 20(16 \cdot \cdot) + 80$. $\cdot^* = 10$, $L^* = 6$, $c^* = \$200$

LL-2:

$\cdot^* = 9.33$, $L^* = 6.67$, $c^* = \$360$. Works MORE ($w \cdot \cdot \cdot^*$ but $L^* = T \cdot \cdot^*$ also $\cdot$ for $\cdot = \cdot = 0.5$)

LL-3:

$\cdot^* = 4$, $L^* = 4$, $c^* = \$100$

LL-4:

$\cdot^* = 4.8$, $L^* = 3.2$, $c^* = \$120$. Yes $\cdot$ more leisure

LL-5:

$\cdot^* = 15.5$, $L^* = 8.5$, $c^* = \$187.50$

LL-6:

(a) $\cdot^* = 6$, $L^* = 4$ (b) $\cdot^* = 7.5$, $L^* = 2.5$ (c) Works LESS as w rises